

Utrecht University

Text And Media Analytics - Group Assignment

# Beyond Fact-Checking: A Comparative Analysis of Agreement, Sentiment, and Thematic Patterns in PolitiFact and Snopes

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**Date: 03/02/2025**

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# Introduction

The rapid spread of misinformation in the digital age has made fact-checking organizations essential in verifying claims and curbing false information. Platforms like PolitiFact and Snopes play a crucial role in this effort, yet their effectiveness and consistency in evaluating claims have come under scrutiny. With Meta's decision to scale back its third-party fact-checking initiatives, concerns have emerged regarding the reliability, agreement, and methodologies used by these independent fact-checking platforms. While their ratings aim to provide clarity, inconsistencies often arise due to differences in rating systems, contextual interpretations, and linguistic nuances. A key challenge in misinformation assessment is evaluating claims that contain high toxicity or strong sentiment, as emotionally charged language may influence how fact-checkers interpret and classify information. This study seeks to answer the main research question:

"What are the factors affecting fact-checking evaluations between PolitiFact and Snopes, particularly focusing on toxicity, sentiment, and thematic patterns."

To answer the research question, we will be looking into two sub-questions:

1. How do toxicity and sentiment levels influence the ratings assigned by PolitiFact and Snopes?
2. What thematic patterns are common across "Fake" and "Real" claims as rated by PolitiFact and Snopes?

Through statistical analysis and linguistic evaluation, this study examines trends in misinformation assessment using a computational approach to fact-checking methodologies. Sentiment and toxicity analysis will be used to evaluate the role of emotionally charged content, and thematic patterns will highlight recurring themes and structures in misinformation. By integrating multiple methods, the study aims to clarify fact-checking transparency, consistency, and potential biases in claim appraisal.

# Literature Review

## Similarity and Differences Between PolitiFact and Snopes

Fact-checking groups include PolitiFact and Snopes (PolitiFact, n.d.; Snopes, n.d.). Identifying various claims and evaluating their veracity, accuracy, and any biases that could affect how information is disseminated and interpreted is the main goal of both PolitiFact and Snopes. Both of them are members of the International Fact-Checking Network (IFCN), a consortium that unites different fact-checking groups under a common umbrella. Five fundamental criteria are upheld by fact-checking organisations thanks to IFCN (IFCN, 2024):

1. Non-partisanship and Fairness
2. Transparency of Sources
3. Transparency of Funding and Organization
4. Transparency of Methodology
5. Open and Honest Corrections Policy

By adhering to these principles, PolitiFact and Snopes aim to maintain credibility and trust in their fact-checking efforts. However, some differences occur in the types of claims both parties assess and the way they rate the claims. Snopes, for example, employs a larger rating system with 20 categories compared to PolitiFact with only six (Snopes, 2024; PolitiFact, 2024). A comprehensive table that summarises the various rating systems is located in Appendix A, Sections 1 and 2. As a result, the disparity in granularity draws attention to the two organisations' disparate scopes. Snopes' more comprehensive and sophisticated rating system implies that its main goal is to look into claims that are true, making sure that those that have been confirmed to be authentic are appropriately categorised. On the other hand, PolitiFact's more streamlined rating system indicates an emphasis on investigating suspicious claims, simplifying its scale to focus on degrees of falsehood rather than nuanced truthfulness.

## Misinformation on Public Opinion and Behavior

The influence of misinformation on public opinion and behavior is very serious. For example, Lazer et al. (2018) show how misinformation about democratic processes may undermine trust in the democratic process through the spread of false claims about how elections and the media work.

This may also affect public health behavior in that in the COVID-19 pandemic spread of fake news regarding health guidelines and vaccination policies has caused a lot of hesitancy, especially among some groups. A peer-reviewed article by Friggeri et al. (2014) discussed how

false health information has rapidly spread across social media, feeding into a broad-based misunderstanding of actual facts that gets in the way of public health interventions.

In this respect, misinformation does not only touch on specific political or health behaviors but also covers general social behavior, such as consumer choice or the enhancement of social cleavages. The ability of misinformation to shape behavior points out the demand for sound information and the elaboration of strategies in contrasting fake news, including improving media literacy and strengthening information verification processes. Vosoughi et al. (2018).

## Sub-Question 1: How do toxicity and sentiment levels correlate with the ratings assigned by PolitiFact and Snopes?

According to research, statements that are regarded as true are typically more neutral or even positive, but those that are rated as false frequently contain more poisonous language and negative sentiment. Venkit and Wilson (2021), for example, discovered that common sentiment and toxicity models assign higher toxicity scores to language that can be associated with misinformation, indicating that erroneous assertions tend to use more forceful or emotionally charged vocabulary. Parallel to this, Venkit et al. (2023) found that biases in these tools are highly associated with poorer truth ratings, indicating that the wording of a claim might influence fact-checkers' perceptions of its correctness. Furthermore, it has been demonstrated by Brassard-Gourdeau and Khoury (2020) that sentiment analysis can be used to identify harmful words at an early stage, which in turn might flag misleading claims before they spread. By turning the fact-checkers' ratings into numbers, researchers can better study how changes in toxicity and sentiment match up with the ratings, offering a clearer picture of the link between language style and bias in claims.

## Sub-Question 2: Thematic Patterns

Fake news is a broad and adaptable phenomenon that can emerge from various sources and be applied to virtually any topic. Its features and structure, however, exhibit recognisable trends. According to a meta-analysis by Tandoc et al. (2020), fake news is a typology with six different categories that are frequently seen in scientific literature rather than a single concept:

- Satire
- Parody
- Fabrication
- Manipulation
- Propaganda
- Advertising

The intricacy of fake news is demonstrated by these six categories, which also highlight how crucial context is in differentiating between fake and authentic news. As demonstrated by the distinctions between satire, propaganda, and blatant fabrication, misinformation can take many different forms, necessitating critical analysis to determine its impact and aim.
A thorough meta-analysis of the common themes, feelings, forms, and objectives of false news is carried out in another paper by Pérez-Escolar et al. (2023).
Pérez-Escolar's analysis of different types of fake news mostly overlaps with that of Tandoc et al. except for that Pérez-Escolar uses categories created by Wardle and Derakhshan (2017) which expands the topology to eight categories instead of the six found in Tandoc et al.:

- Misleading
- False
- Fabricated
- Manipulated
- Impostor
- False
- Satire or parody
- No specific type

This expanded classification reinforces the complexity of fake news, illustrating how different forms of misinformation exploit trust, authority, and context to mislead audiences.

The paper also lists twelve different themes within fake news (with number 1 being the most common theme and number 12 being the least common):

1. Health
2. Politics/democracy
3. Scientific experiment
4. Science (conspiracy theory)
5. Economics, development, and business
6. Education
7. Climate change
8. Immigration
9. Gender
10. Historical facts
11. Famous people
12. Young generation

The importance of a thematic analysis is to understand the different topics and themes that are most prevalent within fake news. This aids in forecasting which misleading stories are most likely to surface and proliferate. Researchers may better understand how disinformation spreads and which subjects are most susceptible to manipulation by charting the thematic landscape of

fake news. This will help them develop more effective measures to lessen the negative effects of fake news on society.

# Data & Methodology

## Dataset Description

The datasets used in this study is derived from a research paper conducted by Harvard Kennedy School researching the behaviour and agreement levels among four different fact-checking organizations. The four organizations researched were Snopes, PolitiFact, Logically, and the Australian Associated Press FactCheck (AAP). The data mainly consists of the articles that are written by the fact-checking organizations, and was gathered using a web scraping algorithm that spans the dates January 1, 2016, to August 31, 2022. Table 1 shows the number of articles scraped for each organization:

| Organization | Number of Articles |
|---|---|
| Snopes | 11639 |
| PolitiFact | 10710 |
| Logically | 4365 |
| AAP | 844 |

**Table 1** - Overview of the dataset sizes

Due to the significant disparity in the number of articles scraped from each organization within the given time frame, this case study will focus solely on PolitiFact and Snopes. These organizations were selected not only for their larger datasets but also to ensure a balanced, consistent, and comprehensive analysis. Table 2 outlines the overlapping attributes present in both datasets:

| Attribute | Explanation |
|---|---|
| `link` | The link to the article of the fact-checked claim |
| `claim` | The claim itself |
| `rating` | A rating for how truthful a claim is |
| `title` | The title of the article written by the fact-checking organization |
| `cdate`/`date_published` | The date of when the article was published |
| `tags` | Tags representing the topics of the article |
| `bodyt` | The body of the article providing an explanation as to how truthful the claim is |
| `sources_num` | The number of sources used to back up the article |
| `sources` | The actual sources |

**Table 2** - Overview of the main attributes in Snopes and PolitiFact datasets

This analysis will focus on the attributes `claim`, `rating`, `bodyt`, and `cdate`/`date_published`. The two organizations also differ in their rating systems and scales, with PolitiFact using six categories and Snopes using fourteen. This difference reflects the varying scopes of the organizations: PolitiFact focuses on verifying suspicious claims, while Snopes places greater emphasis on verifying truthful claims. To ensure that both datasets can be compared properly, the ratings need to be normalized. Table 3 displays the ratings from both organizations and their new normalized rating. An explanation for why certain ratings are either grouped together or removed is given in Appendix A, Section 1.

| Politifact rating | Snopes rating | Normalized Rating |
|---|---|---|
| `TRUE` | `True`, `Correct Attribution`, `Legit`, `Recall` | `True` |
| `mostly-true` | `Mostly True` | `Mostly True` |
| `half-true` | `Mixture` | `Half true` |
| `barely-true` | `Mostly False`, `Miscaptioned`, `Misattributed`, `Outdated` | `Mostly False` |
| `FALSE`, `pants-fire` | `False`, `Scam`, `Originated as Satire`, `Unfounded` | `False` |
| NaN | 'Legend', 'Unproven', 'Labeled Satire', 'Research In Progress', 'No Rating' | REMOVED |

**Table 3** - Mapping of PolitiFact and Snopes Ratings to a Normalized Truthfulness Scale

# Methodology

To answer the research question, the following methods will be employed. To start, an Exploratory Data Analysis (EDA) will be conducted. This step will look at the distribution of the articles across the two organizations and across the normalized rating, to get a preliminary understanding of how balanced the data is. Additionally, the claims and the article bodies (`bodyt`) will be analyzed to visualize the most common words. This helps develop an understanding of the dominating narratives and themes in the dataset, as well as potential biases or points of interest of each organization.

Next, Data Preprocessing will be applied to prepare the data for analysis. This involves calculating TF-IDF vectors for each document (i.e. each claim) and then applying cosine similarity to the vectors to find documents with a similarity score threshold of 0.5 and higher. A similarity score of 0.5 to is chosen to align with the methodology of the paper on which this case study is based. Articles with a rating of 0.5 and higher will be normalized, and below will be removed. Details on the ratings removed and normalized can be found in Table 3.

For further simplification, Feature Engineering is used to further normalize the ratings to a binary classification of 'Real' and 'Fake' called the 'Veracity Score'. This binary classification will make it easier to compare the results across the dataset.

Following EDA, Data Preprocessing, and Feature Engineering, manual assessment will be conducted to evaluate the accuracy, precision, and recall of the matches identified with a similarity score threshold of 0.5. To do this, a random sample containing 50% of the matched claims is manually assessed by four individuals who will rate the match as either 'True', 'Partial' or 'False'. Partial matches are matched claims that need to be further assessed by others due to nuanced differences in the match. See Table 5 for examples. The result of this can then be used to aggregate and make assumptions for all matches, giving us insights into what percentage of the matches are false matches, and at which threshold value (i.e. how accurate the matches are and how reliable the analysis is).

The final step involves Data Analysis, which will be used to answer the main and sub-research questions:.

1. Sub-question 1 - *How do toxicity and sentiment levels correlate with the ratings assigned by PolitiFact and Snopes?*
   Sentiment and toxicity which will be calculated using the libraries Detoxify and `VADER` sentiment analyzer from the `nltk` library. The toxicity and sentiment scores will be calculated for the body text (see table 8 for information on which attribute 'body' refers to) as it provides more context. Using these scores we can answer the second sub-question by developing a correlation heatmap of the toxicity, sentiment, and normalized ratings. Since the normalized ratings are non-numeric ordinal categories, they will be converted into numeric ordinal values to enable correlation analysis.
2. Sub-question 2 - *What are the thematic patterns in the claims categorized as Real or Fake by PolitiFact and Snopes?*
   K-means clustering and Linear Discriminant Analysis (LDA) will be used for topic modelling to identify different themes and topics between Fake and Real claims.

# Results

Below, we have outlined the results and findings for the general research question, as well as for each of the subquestions. The full collection of diagrams and charts are available in the respective appendices.

## Exploratory Data Analysis

### General EDA

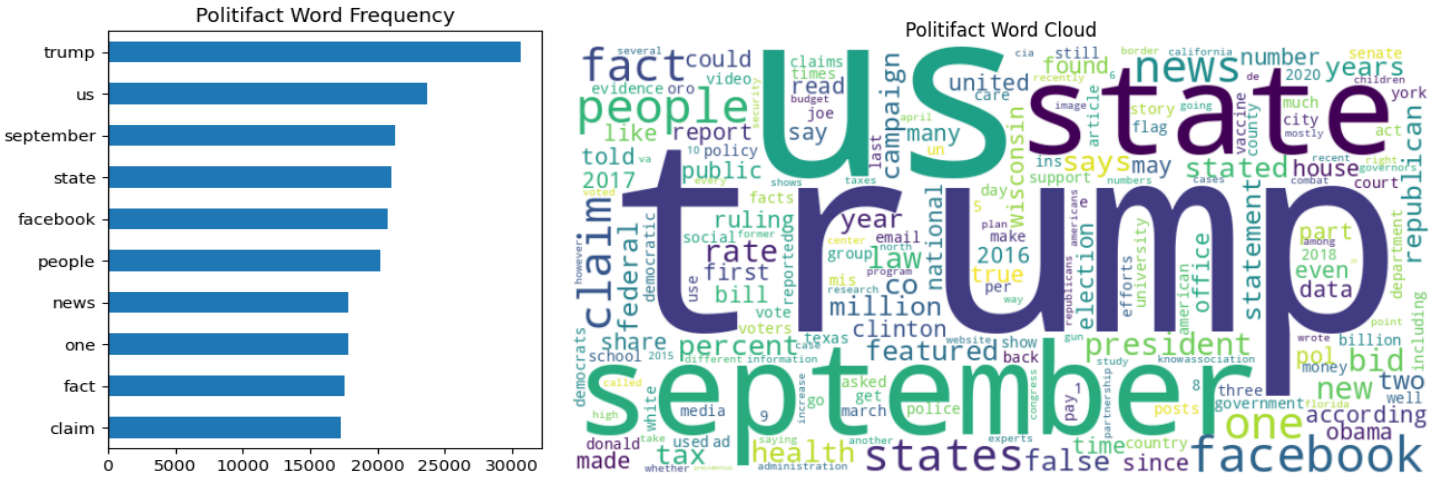



**Figure 1** - PolitiFact Claims Word Frequency Bar Chart and Word Cloud

The most common terms in PolitiFact fact-checked assertions are displayed in Figure 1. Terms like "Trump," "US," and "Facebook" appear often, indicating that a sizable percentage of the fact-checked assertions deal with political and social media-related issues. This suggests that political speech and disinformation are closely related.

The most frequently used terms in Snopes fact-checked claims are shown in Figure 2. Well-known terms like "Trump," "news," "people," and "media" imply that political personalities and popular news subjects are frequently the focus of false information.

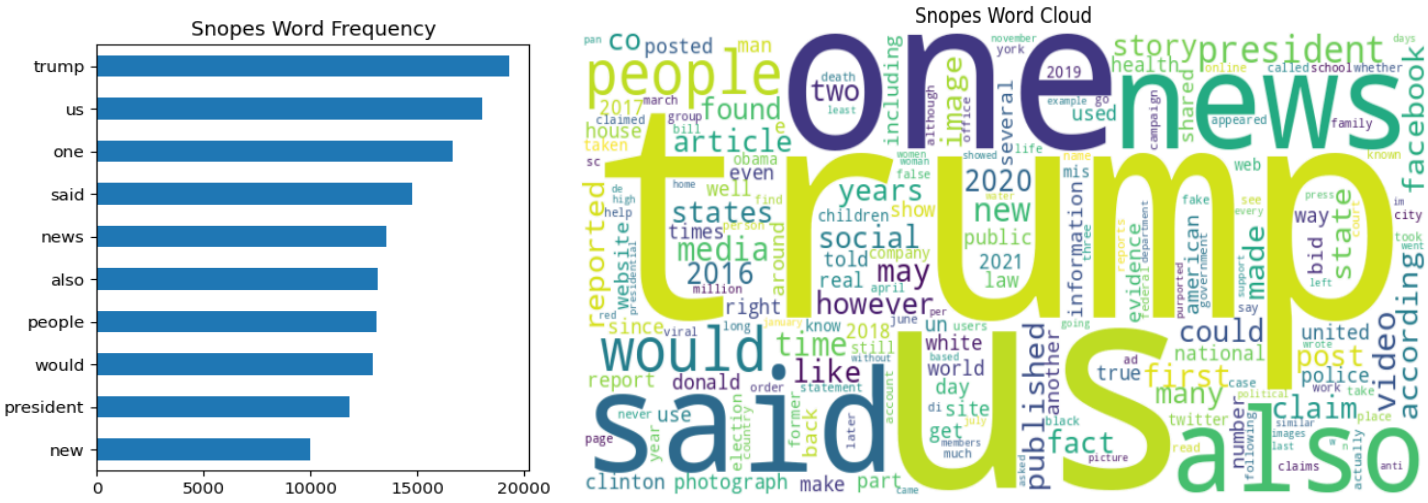



**Figure 2** - Snopes Claims Word Frequency Bar Chart and Word Cloud

The percentage of assertions that Snopes and PolitiFact evaluate as Fake, Half True, or Real is depicted in Figure 3. The fact that a majority of the claims are labeled as Fake emphasizes how common false information is in public conversation. The fact-checking groups' emphasis on identifying and disproving erroneous material is further supported by the limited fraction of assertions that were classified as Real.

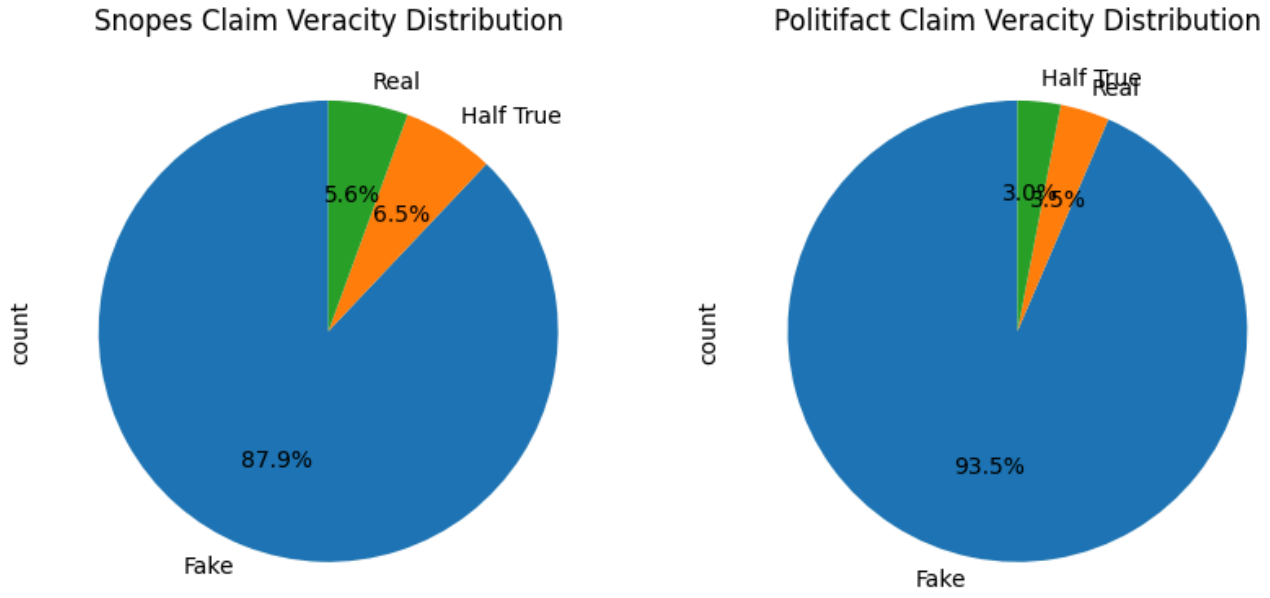



**Figure 3** - PolitiFact and Snopes Pie Charts of Claim Veracity

## Agreement Between Snopes, PolitiFact and Their Ratings

The visualization in Figure 4 compares how frequently Snopes and PolitiFact assigned the same or different ratings to similar claims. The most common match is False → False, indicating strong agreement between both fact-checking platforms in detecting false claims. Discrepancies in ratings suggest differences in evaluation criteria or context-dependent fact-checking methodologies.

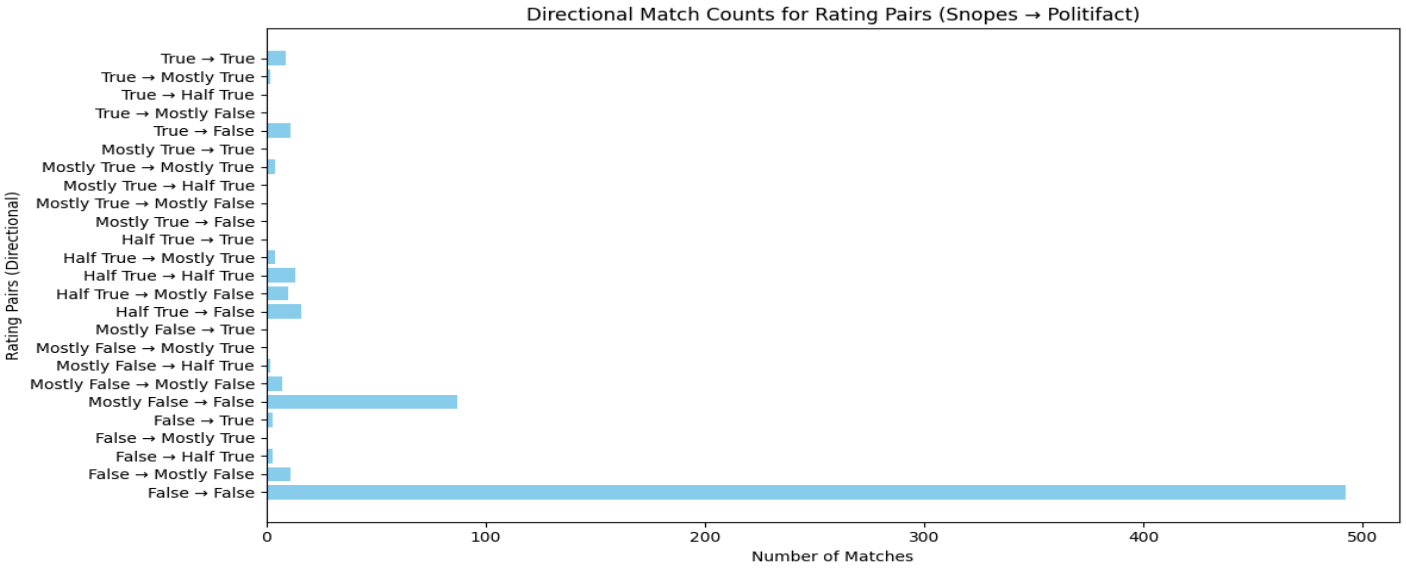



**Figure 4** - Bar Chart of Veracity Rating Pair Matches Between Snopes and PolitiFact

## Matching Accuracy

To check the accuracy of the TF-IDF matching a random sample of size 50% was taken and manually assessed and the results were aggregated to the entire dataset of matched claims. Based on the manual assessment, 83.86% were true positives, 11.53% were false positives, and 4.61% were partial positives.

| TP/FP/PP | Count | Percentage |
|---|---|---|
| TP | 291 | 83.86% |
| FP | 40 | 11.53% |
| PP | 16 | 4.61% |

**Table 4** - Proportion of True Positives, False Positives and Partial Positives in the dataset. The 'Count' reflects a sample size of 50%.

Partial positives (PP) are matches that had some overlap but differed in some ways that made matching it to either TP or FP difficult. For example, some claims were talking about the same claim but one of the claims missed a certain detail or had a slightly different value that changed the way the fact-checker would go about rating the claim (Table 5, Row 0). Another example in Row 1 looks at the same claim except that the Snopes claim analyses a broader version of the same claim while PolitiFact looks at a specific aspect of the claim; i.e., black unemployment vs overall unemployment. A third example involves two claims that are talking about very similar events but at different periods and different contexts. Table 5, row 2 talks about a memorial that was vandalized during a black lives matter protest, except, when looking at the context of the article the claims are referring to different memorials and different black lives matter protests.

| | Snopes Claim | PolitiFact Claim |
|---|---|---|
| 0 | The crew of the USS Constitution ("Old Ironsides") consumed more than 250,000 gallons of liquor and no water at all during a six-month voyage in 1798. | The crew of the U.S.S. Constitution consumed more than 252,000 gallons of alcohol and only 10,000 gallons of water on a seven-month trip. |
| 1 | Because of U.S. President Donald Trump, America recorded the lowest unemployment rate in 50 years. | "Because of my policies, Black Unemployment has just been reported to be at the LOWEST RATE EVER RECORDED!" |
| 2 | The 9/11 Memorial & Museum and the World War II Memorial, both in New York state, were vandalized during Black Lives Matter protests in the spring of 2020. | Says a photo shows a memorial vandalized by Black Lives Matter protesters. |

**Table 5** - Examples of Partial Positive Matches From a Random Sample of the Dataset.

The proportions of TP, FP, and PP were then analyzed at different TF-IDF threshold values to see which threshold values contained the least FPs, shown in Figure 5. At a threshold of 0.649 there are close to 300 data points left with virtually no FPs and only a few PPs. However, due to the sharp decrease in data points for each threshold value, after starting with close to 12000 before matching and around 700 after, the original threshold of 0.5 was kept. This analysis does show, however, that the accuracy of the matching should be taken into account when answering the research question.

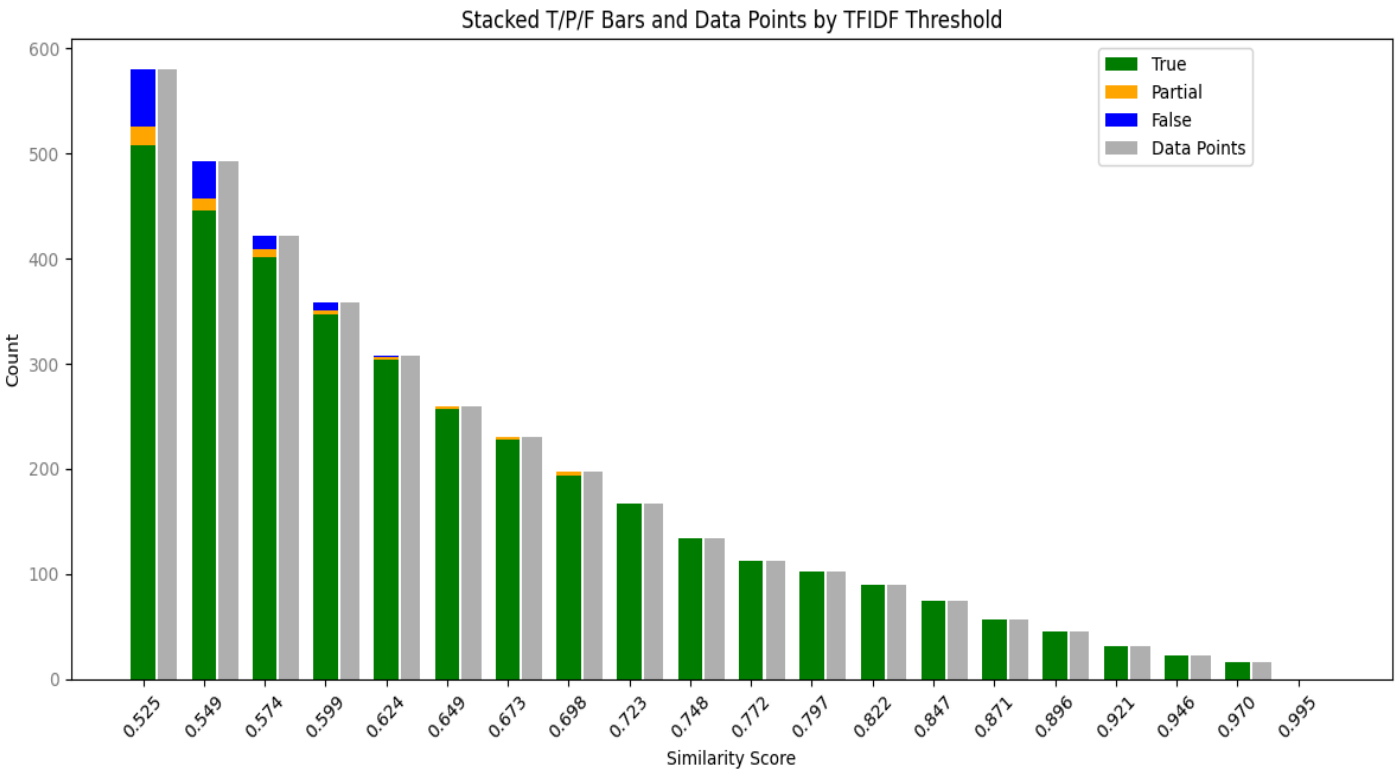



**Figure 5** - Proportion of True Positive and False Positive Matched Claims by Similarity Score.

## Sub-question 1: How do toxicity and sentiment levels correlate with the ratings assigned by PolitiFact and Snopes?

The analysis of the visual data reveals several consistent patterns in how toxicity and sentiment levels relate to fact-checking ratings provided by Snopes and PolitiFact. Figure 6 shows that both sources exhibit a marked concentration of low toxicity scores overall, with the majority of claims registering a toxicity score of 0 for both datasets. This similarity suggests that most claims tend to have non-toxic language; however, a noticeable minority of false and mostly false claims show elevated toxicity levels.

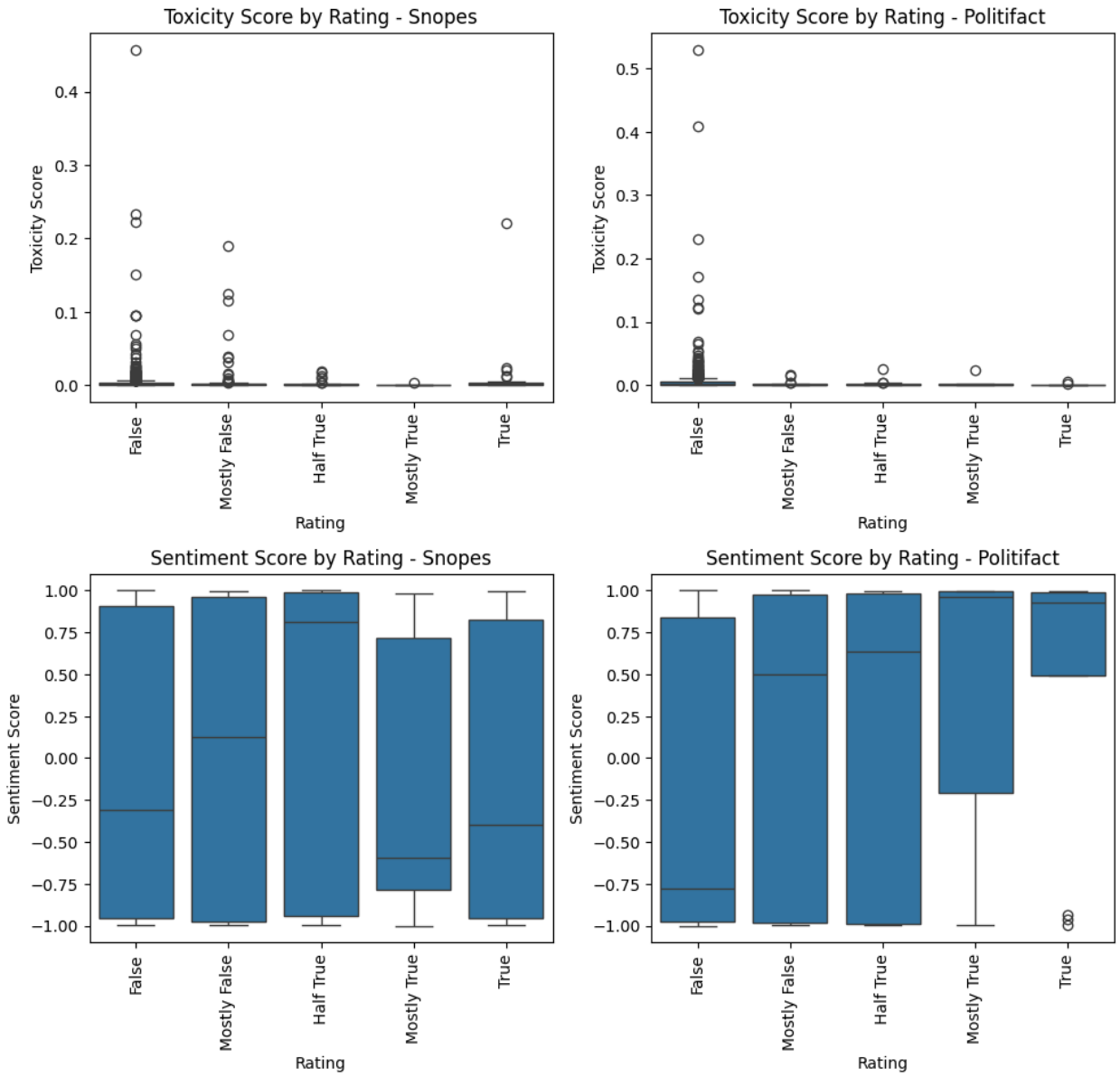



**Figure 6** - Toxicity and Sentiment Score Box Plot Across Ratings

In contrast, the sentiment distributions for both platforms in Figure 7 are bimodal, with peaks observed at the extremes of the sentiment scale, indicating that the claims tend to have either very negative or very positive sentiment. The box plots in Figure 6, that break down sentiment by rating category, further reinforce this observation. For both Snopes and PolitiFact, claims rated as “False” or “Mostly False” predominantly exhibit negative sentiment scores, while claims deemed “True” or “Mostly True” tend to have positive sentiment scores. The “Half True” category occupies a more neutral position. These results imply a correlation between the perceived veracity of a claim and its emotional tone: the more untruthful a claim is, the more negative its sentiment tends to be, and vice versa.

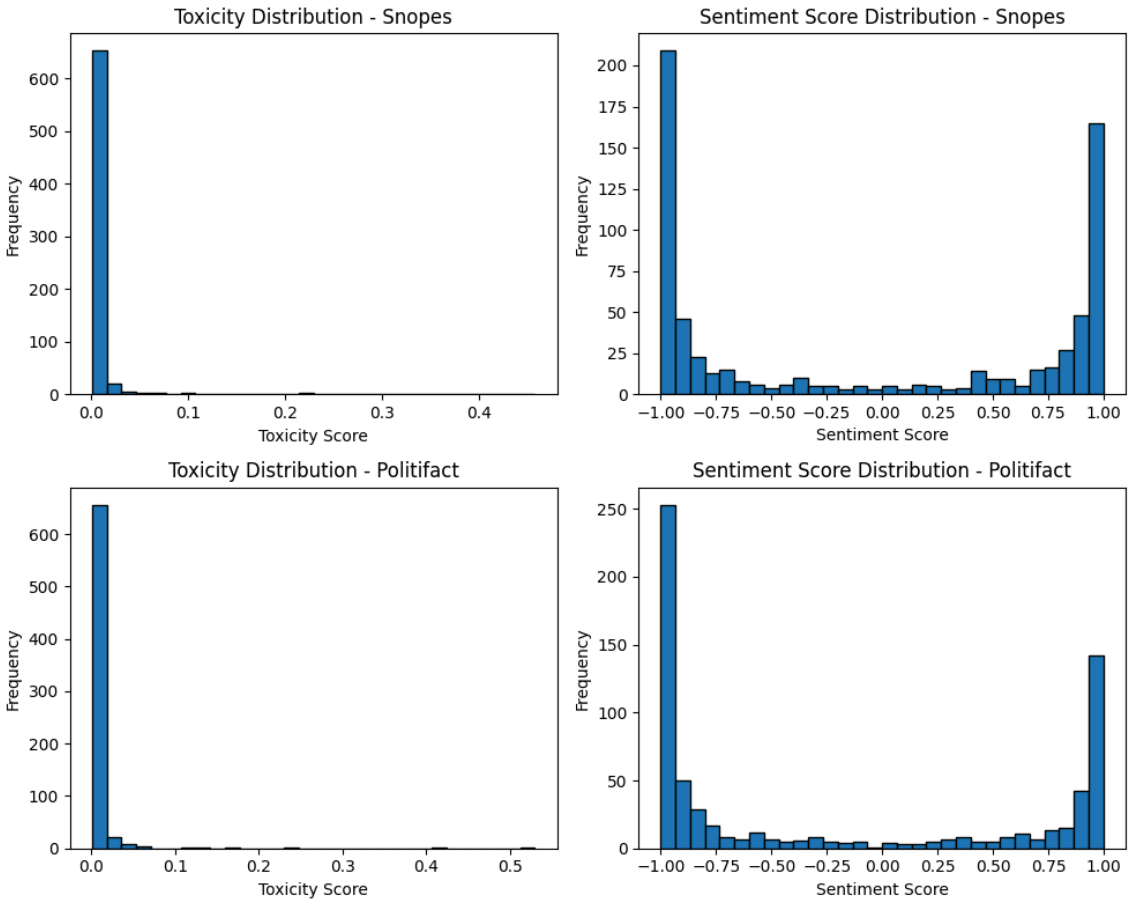



**Figure 7** - Toxicity and Sentiment Score Distribution

The correlation heatmap in Figure 8 shows that toxicity scores from Snopes and Politifact are moderately positively correlated (r = 0.55). Moreover, numeric ratings from both sources correlate significantly (r = 0.54), while sentiment scores between the two sources show a moderate positive association (r = 0.47). Despite potential differences in methodologies and source-specific nuances, the measures of toxicity and sentiment—and the corresponding truth ratings—tend to align across fact-checking organizations.

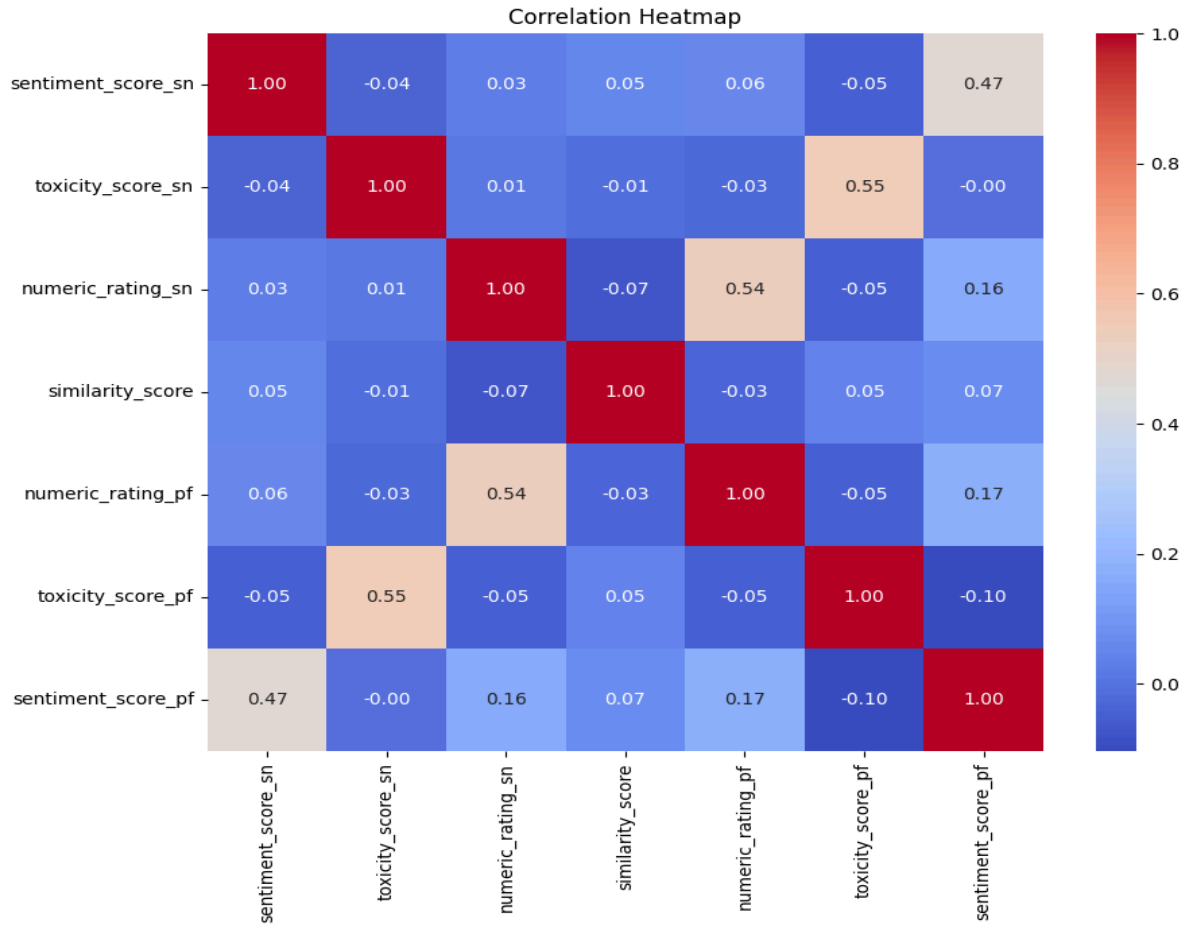



**Figure 8** - Correlation heatmap between various numerical variables of Snopes and Politifact

## Sub-question 2: What thematic patterns are common across "Fake" and “Real” claims as rated by PolitiFact and Snopes?

| Rank | Snopes - Real | | Snopes - Fake | | PolitFact - Real | | PolitiFact - Fake | |
|---|---|---|---|---|---|---|---|---|
| | Topic | **Count** | Topic | **Count** | Topic | **Count** | Topic | **Count** |
| **1** | 5 | **12** | 6 | **154** | 13 | **7** | 0 | **186** |
| **2** | 2 | **7** | 0 | **130** | 12 | **6** | 3 | **71** |
| **3** | 3, 6 | **5** | 2 | **87** | 0 | **5** | 5 | **64** |

**Table 6** - Top three topics per organization and veracity

Table 6 is a summary table containing the top 3 topics per organization and veracity score, while Appendix C, Section 3 contains all the topic names which have been derived using the top 10 words per topic.

For real claims, Snopes primarily focuses on current events and news reporting, particularly verifying political statements and trending narratives (Topic 5). Additionally, misinformation in media and social networks (Topic 2) and fact-checking of international affairs and statistical verification (Topics 3 and 6) emerge as significant themes. These results suggest that Snopes' fact-checking efforts emphasize debunking claims related to political discourse, public figures, and statistical interpretations.

Snopes' fake claims are dominated by image manipulation and fake news spread (Topic 6), suggesting that a large portion of misinformation relies on visual deception. Other prevalent themes include misinformation in news publishing (Topic 0) and false political quotes on social media (Topic 2). This indicates that misinformation circulates heavily through manipulated images, viral memes, and misleading statements attributed to political figures.

PolitiFact's real claims are centered around federal policies and legislative processes (Topic 13), followed by social media influence on public perception (Topic 12) and digital media fact-checking (Topic 11). This suggests that PolitiFact often verifies government policies, legislative discussions, and the role of social media in spreading or countering misinformation.

According to Snopes' findings on manipulated imagery and viral content, false information spread through visual and social media posts is the primary type of fraudulent claims (Topic 0). Other significant issues include election integrity and political manipulation (Topic 3), as well as fake news headlines and media distortion (Topic 5). These findings suggest that political deception is a widespread problem, particularly when it comes to elections, public figures, and false news reports.

Both PolitiFact and Snopes emphasise fact-checking political and social narratives; however, Snopes appears to focus more on disputing specific news reports and statistical claims, whilst PolitiFact typically addresses misleading information pertaining to legislative debate and government policy. Additionally, visual misrepresentation is a major issue on both platforms, which highlights the part that Photoshopped photos and viral social media material play in.

# Discussion

## Sub-Question 1

The significant inverse relationship between agreement and toxicity draws attention to a crucial problem: provocative language infuses subjectivity into verification procedures in addition to warping factual interpretation. This implies that when evaluating high-toxicity assertions, fact-checkers could be swayed by unconscious biases and contextual subtleties, even using comparable procedures.

The finding that assertions with more negative emotion and toxicity are more likely to be labelled as false by both Politifact and Snopes further reflects these difficulties. The glaring bimodal distribution of mood and the consistently low toxicity of the majority of assertions point to the need for more sophisticated fact-checking techniques that can take into consideration the biases that are inherently introduced by emotionally charged language.

### Limitations

The dataset we used comes from a Harvard article that merges fact-checking data from Snopes and PolitiFact. Originally, claims were matched based on the Claims column, which contains short statements like "Donald Trump did this." But this method missed matches due to wording differences. We also attempted to match the claims using the Body column, however this produced a maximum TF-IDF score of about 0.8 and many more errors. Because TF-IDF reduces text to a bag of words and eliminates context, it is not a good method for matching.

Moreover, for sentiment and toxicity analysis the Body columns were used instead of the claims as the claims were usually one sentence, which is too short to derive any meaningful toxicity or sentiment from. Using the Body introduces problems such as including additional sentiment or toxicity as the column not only summarizes the article but also includes quotes and the fact-checker's remark. Sentiment and toxicity level analyses may be impacted since these factors may represent the viewpoint of the fact-checker rather than the tone of the original assertion. As a result, although this method increases matching accuracy, it also requires considerable thought.

### Future Research

It is advised that contextual word embeddings be used in future studies when comparing claims based on the Body column. Because the claims were similar in wording but distinct in context, many of the matches were incomplete or incorrect matches, as the matching accuracy section demonstrates. This issue might be resolved and more accurate matches could be obtained by using contextual word embeddings like ELMo, BERT, or GPTs.

## Sub-Question 2

The majority of phoney claims on PolitiFact and Snopes include politically framed content, viral social media posts, and altered photographs, especially when it comes to elections, public figures, and governmental policy. The lack of information on real claims that was acquired throughout the matching process, on the other hand, makes them seem more dispersed and less thematically unified. Additionally, Topic 6 in Snopes Fake and Topic 0 in PolitiFact Fake highlight how visual deception is a major factor in the spread of fake news. This implies that since visual misinformation is more contagious and people are more likely to believe information that they can see, fact-checking initiatives should concentrate more on image and video verification (Yang et al., 2023).

These findings are consistent with previous research on fake news typologies and thematic structures. The high occurrence of political and election-related misinformation aligns with the categorization of fake news by Tandoc et al. (2020), which identified propaganda, fabrication, and manipulation as major types of misinformation. Additionally, Pérez-Escolar et al. (2023) found that health, politics, and conspiracy theories are the dominant themes in fake news, which is similar to the topics that were assimilated using Snopes and PolitiFact's datasets.

### Limitations

Despite these insights, researching thematic patterns in fact-checking presents several limitations. During the EDA, it was shown that there was a noticeable imbalance in the amount of real and fake claims that was scraped from PolitiFact and Snopes (Figure 5). This made it harder to establish clear overarching topics for real claims. Many of Politifact’s topics contain very few claims, making it challenging to identify common patterns. The skewed veracity is logical in organizations that are established to identify misinformation as most claims they would consider to look into are suspicious to begin with and, therefore, most likely false.

### Future Research

For future research, it would be beneficial to add more depth to topic assignments in order to allow for more accurate conclusions to be made about the thematic structure of fact-checking efforts. Moreover, fact-checking organizations should prioritize addressing visual misinformation, given the dominance of manipulated images and social media posts in fake claims.

## Conclusion

We discovered that, in contrast to genuine claims, false claims typically display greater toxicity and more negative sentiment. Despite these variations, most assertions were marked as "Fake," highlighting the fact-checkers' focus on dispelling false information. Even with a rigorous similarity level (0.5), 11% of matches were false positives, and minor language or context differences (e.g., "6 months" versus "7 months") occasionally resulted in rating discrepancies. Furthermore, our thematic analysis showed that modified images and politically charged social media posts are frequently at the heart of false assertions, while real claims were more fragmented, typically focusing on policy debates.

In general, our results are in line with earlier investigations, particularly Tandoc et al. (2020) and Pérez-Escolar et al. (2023). However, political narratives and visual deception were given more weight in our data, which was gathered from general fact-checkers. Furthermore, our moderate correlation ($r = 0.55$) suggests that fact-checkers' techniques are reasonably resistant to the influence of emotional language, despite previous research suggesting that toxic language may bias fact-checkers (Venkit and Wilson, 2021; Venkit et al., 2023; Brassard-Gourdeau & Khoury, 2020). However, the small proportion of "Real" claims in our dataset points to a knowledge gap about factual information, emphasising the need for more impartial data collection and improved instruments to better capture the nuances of visual misleading in future studies.

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# Appendix A - Literature Review

## Section 1 - Snopes Fact-Checking Rating Summary

| Rating | Explanation |
|---|---|
| Research in Progress | This rating is used with pages that present items which we are currently investigating but have not yet reached any publishable conclusion about. |
| True | This rating indicates that the primary elements of a claim are demonstrably true. |
| Mostly True | This rating indicates that the primary elements of a claim are demonstrably true, but some of the ancillary details surrounding the claim may be inaccurate. |
| Mixture | This rating indicates that a claim has significant elements of both truth and falsity to it such that it could not fairly be described by any other rating. |
| Mostly False | This rating indicates that the primary elements of a claim are demonstrably false, but some of the ancillary details surrounding the claim may be accurate. |
| False | This rating indicates that the primary elements of a claim are demonstrably false. |
| Unproven | This rating applies to a claim for which we have examined the available evidence but could not arrive at a true or false determination, meaning the evidence is inconclusive or self-contradictory. |
| Unfounded | This rating applies to a claim that we have investigated but found no demonstrable evidence to support it. Such claims typically originate as hearsay, speculation, or groundless rumor. |
| Outdated | This rating applies to items for which |

| | |
|---|---|
| | subsequent events have rendered their original truth rating irrelevant (e.g., a condition that was the subject of protest has been rectified, or the passage of a controversial law has since been repealed). |
| Miscaptioned | This rating is used with photographs and videos that are “real” (i.e., not the product, partially or wholly, of digital manipulation) but are nonetheless misleading because they are accompanied by explanatory material that falsely describes their origin, context, and/or meaning. |
| Correct Attribution | This rating indicates that quoted material (speech or text) has been correctly attributed to the person who spoke or wrote it. |
| Misattributed | This rating indicates that quoted material (speech or text) has been incorrectly attributed to a person who didn't speak or write it. |
| Legend | This rating is most commonly associated with items that describe events so general or lacking in detail that they could have happened to someone, somewhere, at some time, and are therefore essentially unprovable. |
| Scam | This “rating” is not a truth rating but rather indicates pages that describe the details of verified scams. |
| Legit | This rating describes authentic, legitimate offers or initiatives available to consumers or members of the public. |
| Labeled Satire | This rating indicates that a claim is derived from content described by its creator and/or the wider audience as satire. Not all content described by its creator or audience as ‘satire’ necessarily constitutes satire, and this rating does not make a distinction between 'real' satire and content that may not be effectively recognized or understood as satire despite being labeled as such. |
| Originated as Satire | This rating refers to content that originally |

| | |
|---|---|
| | came from a site described as satire, but was later stripped of some of its satirical markings, repackaged, and posted elsewhere. The rating also applies to content not necessarily labeled as satire but that audiences perceived as satirical nonetheless, such as content from The Onion. |
| Recall | This rating is applied to genuine product recall notices. |
| Lost Legend | These legends aren't really lost — we've known where they were the whole time! We created The Repository of Lost Legends (TRoLL for short) for those of you who don't let the truth get in the way of a good story. If you have a taste for the unusual and arcane (and can suspend your disbelief just a little), sample some of these precious gems. |
| Fake | This rating applies to photographs and videos that were either manipulated via digital editing, or manufactured from scratch via CGI (computer-generated imagery) or AI (artificial intelligence) software, such that they convey a false or misleading message or an outright misrepresentation of reality. |

**Table 7** - Rating System used by Snopes (Fact Check Ratings, n.d.)

## Section 2 - PolitiFact Fact-Checking Rating Summary

| Rating | Description |
|---|---|
| **TRUE** | The statement is accurate and there's nothing significant missing. |
| **MOSTLY TRUE** | The statement is accurate but needs clarification or additional information. |
| **HALF TRUE** | The statement is partially accurate but leaves out important details or takes things out of context. |
| **MOSTLY FALSE** | The statement contains an element of truth but ignores critical facts that would give a different impression. |
| **FALSE** | The statement is not accurate. |
| **PANTS ON FIRE** | The statement is not accurate and makes a ridiculous claim. |

**Table 8** - Rating system used by PolitiFact (Holan, n.d.)

# Appendix B - Methodology

## Section 1 - Justification for Rating Normalization

Rationale for Rating Removal and Normalisation

We mapped Snopes and PolitiFact's fact-check scores into a standardized 5-point scale: True, Mostly True, Half True, Mostly False, and False in order to compare them. This maintains significant differences while guaranteeing consistency because the two platforms employ different grading systems

Normalization of Ratings

**True**: Includes PolitiFact's "True" and Snopes' "True," "Correct Attribution," "Legit," and "Recall." These all indicate claims that are factually accurate without major distortions.

**Mostly True**: Covers PolitiFact "Mostly True" and Snopes' equivalent ratings. These claims are largely accurate but may need additional clarification.

**Half True**: Groups PolitiFact's "Half True" with Snopes' "Mixture," both of which describe claims containing a mix of truth and falsehood.

**Mostly False**: Includes PolitiFact's "Barely True" and Snopes' "Mostly False," "Miscaptioned," "Misattributed," and "Outdated." These ratings apply to statements that contain some truth but are largely misleading.

**False**: Merges PolitiFact's "False" and "Pants on Fire" with Snopes' "False," "Scam," "Originated as Satire," and "Unfounded." These ratings cover claims that are entirely inaccurate or deceptive.

This approach smooths out inconsistencies across the datasets while preserving distinctions in truthfulness.

Exclusion of Certain Snopes Ratings

Some Snopes ratings were excluded because they don't assess factual accuracy in a way that fits this scale:

**Legend**: Based on folklore or unverifiable stories rather than verifiable claims.

**Labeled Satire**: Content explicitly identified as satire, meaning it was never intended to be factual.

**Research in Progress**: Represents claims still under investigation, with no definitive conclusion.

**No Rating**: Not applicable for truth assessment.

Excluding these ensures the analysis remains focused on clear truth evaluations, avoiding ambiguity and keeping comparisons between fact-checking platforms as consistent as possible.

# Appendix C - Sub-question 2

## Section 1 Elbow Method

The following graphs in Figure 9 show the use of the elbow method to determine the optimal number of clusters for K-means clustering. K-means clustering will later be used in tandem with LDA to identify more granular topics by reducing the dataset into smaller, more coherent subsets. However, due to a linear decrease in inertia for Snopes and PolitiFact with a veracity score of 'Real', the 'k' value had to be manually chosen based on a visual assessment.

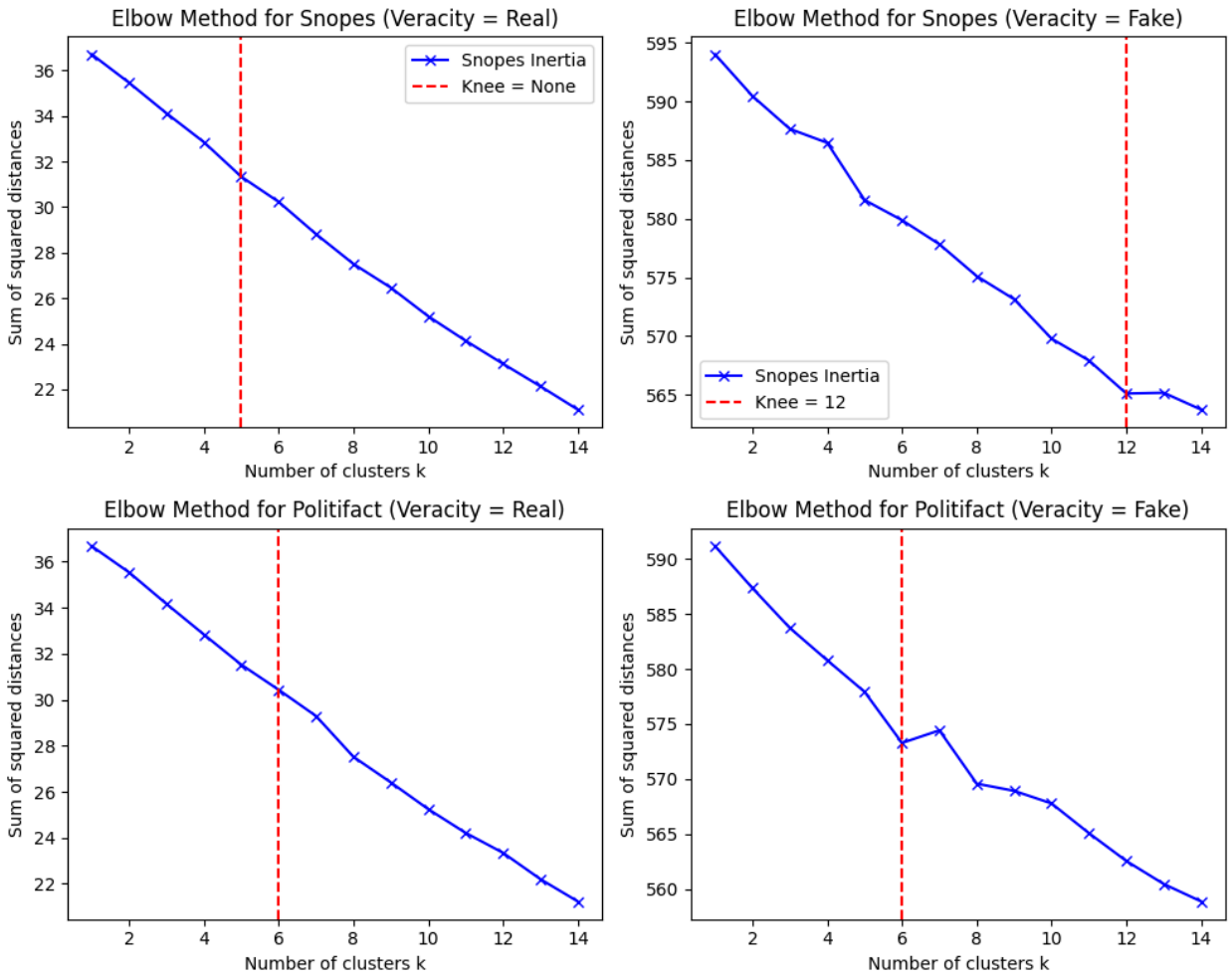



**Figure 9** - Elbow Method for Snopes and Politifact for Veracity = Fake and Veracity = Real

## Section 2 - Cluster Sizes

The table shows the number of documents per cluster size separated into Real and Fake claims for Snopes and PolitiFact. Here `sn` stands for Snopes and `pf` stands for PolitiFact. This table highlights imbalanced clusters, as well as meaningful clusters with more documents than others. It gives an insight into possible themes that dominate Real and Fake claims.

| k_means_sn_real | body_sn_real | | k_means_sn_fake | body_sn_fake | | k_means_pf_real | body_pf_real | | k_means_pf_fake | body_pf_fake |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 7 | | 0 | 49 | | 0 | 2 | | 0 | 175 |
| 1 | 2 | | 1 | 81 | | 1 | 7 | | 1 | 108 |
| 2 | 13 | | 2 | 20 | | 2 | 10 | | 2 | 85 |
| 3 | 7 | | 3 | 33 | | 3 | 4 | | 3 | 127 |
| 4 | 10 | | 4 | 10 | | 4 | 12 | | 4 | 72 |
| - | - | | 5 | 81 | | 5 | 4 | | 5 | 44 |
| - | - | | 6 | 23 | | - | - | | - | - |
| - | - | | 7 | 43 | | - | - | | - | - |
| - | - | | 8 | 33 | | - | - | | - | - |
| - | - | | 9 | 186 | | - | - | | - | - |
| - | - | | 10 | 37 | | - | - | | - | - |
| - | - | | 11 | 15 | | - | - | | - | - |

**Table 9** - number of documents per cluster size in Real and Fake claims across Snopes and PolitiFact

## Section 3 - Top Words and Document Count Per LDA Topic

### Snopes - Real Claims

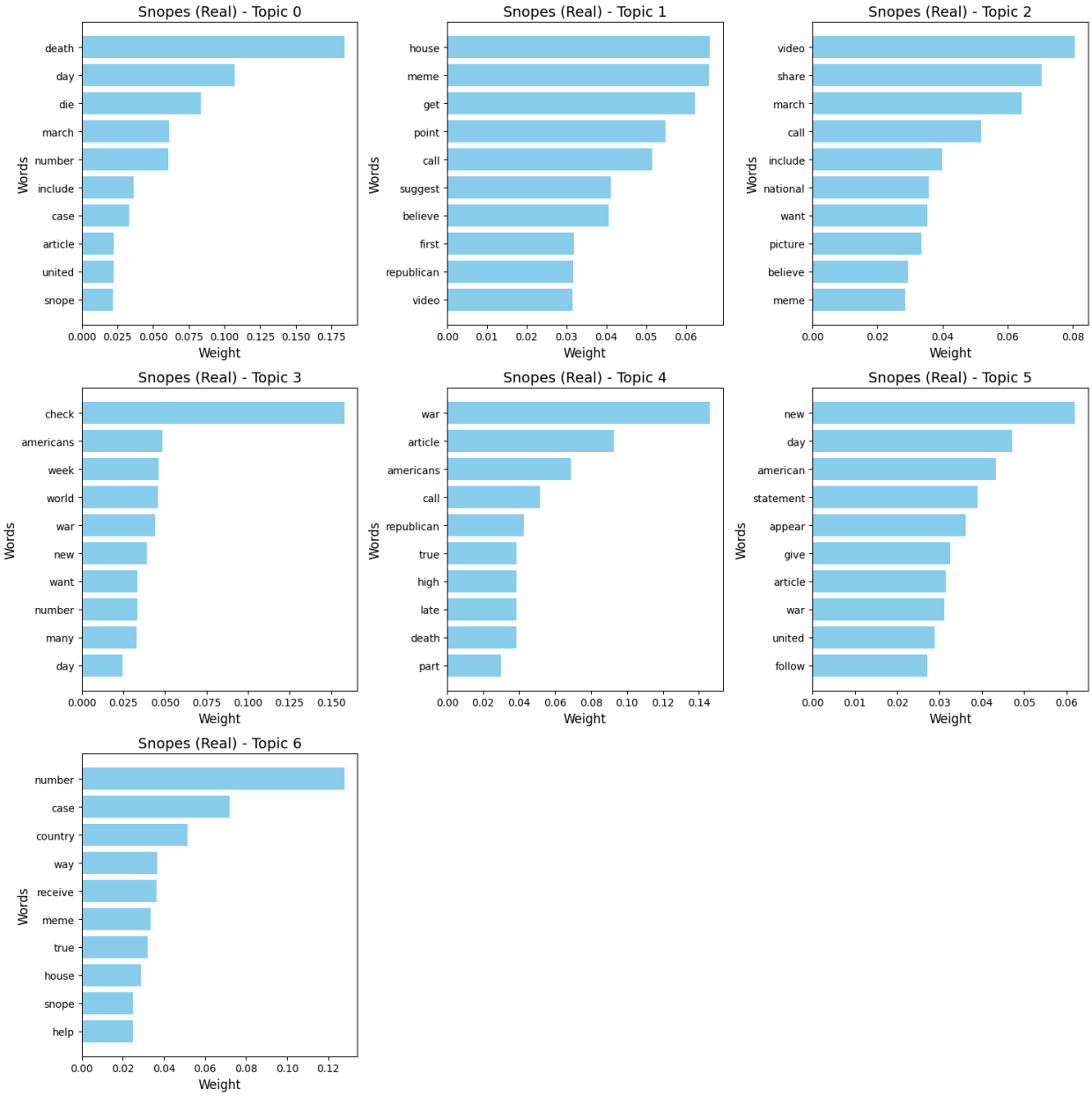



**Figure 10** - Most occurring words per LDA topic in Snopes, Real claims

| Topic Number | Topic Name | Explanation | Document Count |
|---|---|---|---|
| Topic 0 | Mortality and Public Health Events | This topic centers around discussions related to mortality, | 4 |

| | | | |
|---|---|---|---|
| | | possibly concerning major health crises, significant events, or policy responses. | |
| Topic 1 | Political Narratives and Public Discourse | This topic highlights political discussions, belief-driven narratives, and the influence of memes in shaping public perception. | 4 |
| Topic 2 | Misinformation in Media and Social Networks | Focuses on the role of videos and images in misinformation, particularly how media content is shared and interpreted. | 7 |
| Topic 3 | Fact-Checking and International Affairs | Covers fact-checking efforts related to global issues, including war, international relations, and national events. | 5 |
| Topic 4 | Political Statements and Historical Context | Discusses political statements, historical references, and conflicts, possibly involving political parties or leaders. | 2 |
| Topic 5 | Current Events and News Reporting | Covers trending news, statements from public figures, and political developments shaping public discourse. | 12 |
| Topic 6 | Statistical Reports and Verification | Highlights numerical data, case statistics, and verification | 5 |

| | | | |
|---|---|---|---|
| | | efforts, possibly focusing on misinformation correction. | |

**Table 10** - Description of each LDA topic in Snopes, Real claims

Snopes - Fake Claims

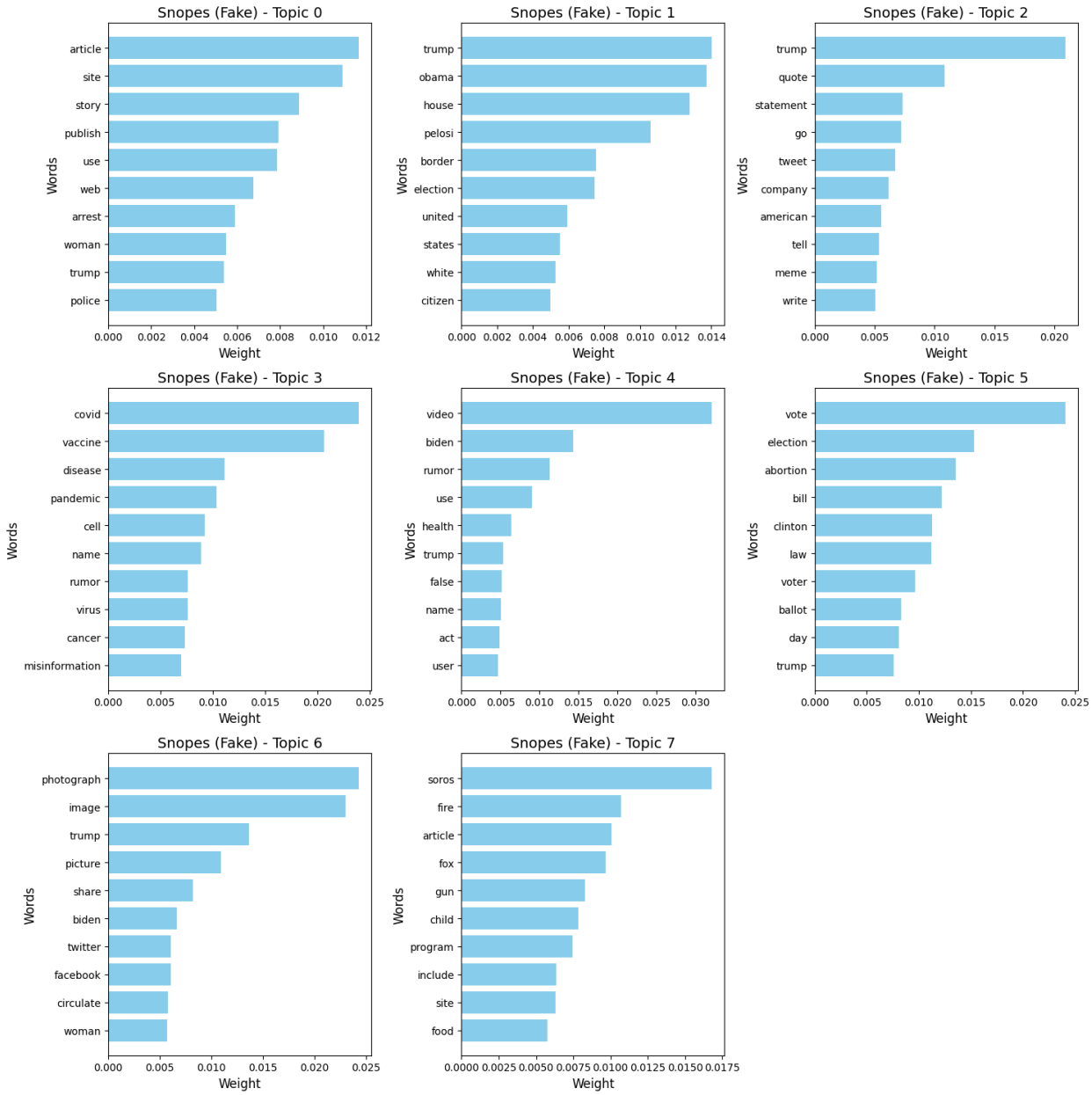



**Figure 11** - Most occurring words per LDA topic in Snopes, Fake claims

| Topic Number | Topic Name | Explanation | Document Count |
|---|---|---|---|
| Topic 0 | Misinformation in News and Publishing | This topic focuses on false claims related to news articles, website sources, and potential misinformation involving crime, arrests, and political figures. | 130 |
| Topic 1 | Political Figures and Immigration Misinformation | Contains misinformation about political leaders (Trump, Obama, Pelosi), border policies, elections, and U.S. citizenship. | 63 |
| Topic 2 | Social Media and Political Quotes | Focuses on fake political quotes, social media posts (tweets, memes), and misinformation about statements attributed to politicians. | 87 |
| Topic 3 | COVID-19 and Vaccine Misinformation | Covers misinformation related to the COVID-19 pandemic, vaccine conspiracies, and health-related falsehoods. | 52 |
| Topic 4 | Health and Election Conspiracies | Misinformation regarding Joe Biden, Trump, health-related conspiracies, and misleading claims about legislation or government acts. | 43 |
| Topic 5 | Election Fraud and Political Legislation | Focuses on false narratives around election fraud, | 44 |

| | | | |
|---|---|---|---|
| | | abortion laws, political bills, and claims about voting procedures. | |
| Topic 6 | Image Manipulation and Fake News Spread | Highlights misinformation spread through manipulated images and fake pictures, often circulated on social media platforms like Twitter and Facebook. | 154 |
| Topic 7 | Conspiracies Involving Influential Figures and Policies | Covers conspiracies about high-profile figures like George Soros, gun control, child-related programs, and misinformation spread by media organizations. | 38 |

**Table 11** - Description of each LDA topic in Snopes, Fake claims

PolitiFact - Real Claims

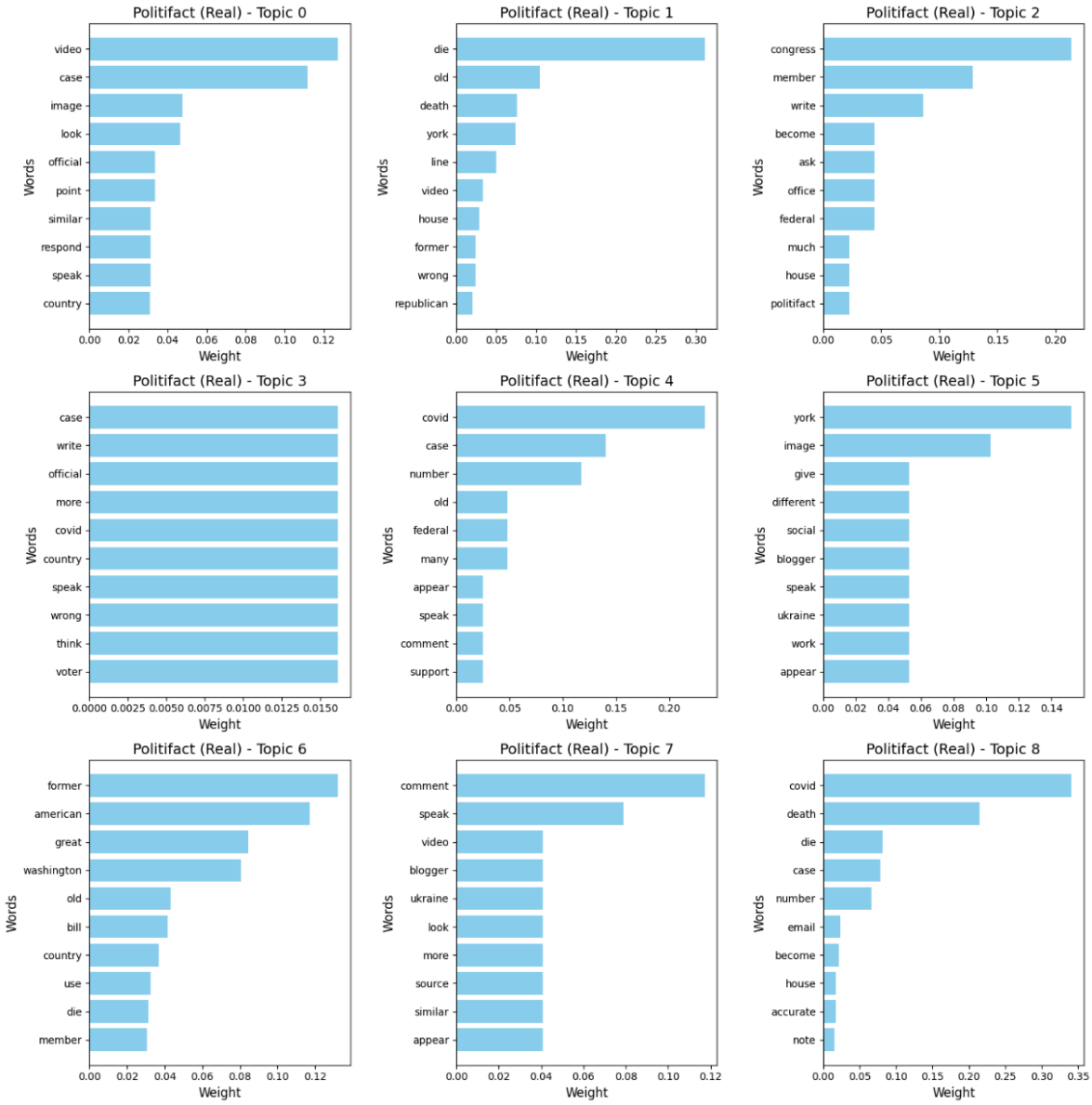



**Figure 12** - Most occurring words per LDA topic in Politifact, Real claims

| Topic Number | Topic Name | Explanation | Document Count |
|---|---|---|---|
| Topic 0 | Media Representation and Visual Misinformation | This topic seems to focus on how media and images are used in misinformation and fact-checking, with emphasis on official statements | 5 |

| | | | |
|---|---|---|---|
| | | and responses. | |
| Topic 1 | Mortality and Demographics in Misinformation | This topic relates to discussions about mortality, age-related claims, and political narratives. | 1 |
| Topic 2 | Political Structures and Government Officials | This topic is about government-related claims, legislative processes, and political office discussions. | 1 |
| Topic 3 | Voting and Election Integrity | Focuses on voter-related misinformation, election processes, and the impact of COVID-19 on elections. | 0 |
| Topic 4 | COVID-19 and Government Response | Discusses misinformation around COVID-19 statistics and government response. | 1 |
| Topic 5 | International Affairs and Social Media Discourse | Covers global topics, particularly Ukraine, war narratives, and the role of social media. | 1 |
| Topic 6 | Historical and Political Figures | Discusses historical and political figures, their influence, and legislation. | 3 |
| Topic 7 | Online Commentary and Public Discourse | Focuses on online conversations, influencers, and how public discourse shapes misinformation. | 1 |

| | | | |
|---|---|---|---|
| Topic 8 | Health and COVID-19 Misinformation | Covers COVID-19 deaths, medical misinformation, and public health discussions. | 3 |
| Topic 9 | Political Institutions and Public Offices | Focuses on political structures, university affiliations, and voter concerns. | 2 |
| Topic 10 | Government Operations and Policy Discussions | Covers government offices, White House policies, and foreign relations. | 1 |
| Topic 11 | Digital Media and Fact-Checking | Discusses how misinformation spreads via social media and fact-checking efforts. | 4 |
| Topic 12 | Social Media and Public Perception | Covers the role of visual media in misinformation, particularly about Ukraine and social networks. | 6 |
| Topic 13 | Federal Policies and Legislative Process | Discusses legislative procedures, policy-making, and governance. | 7 |
| Topic 14 | Education, Policy, and Political Campaigns | Covers university policies, campaign misinformation, and governance-related claims. | 3 |

**Table 12** - Description of each LDA topic in Politifact, Real claims

PolitiFact - Fake Claims

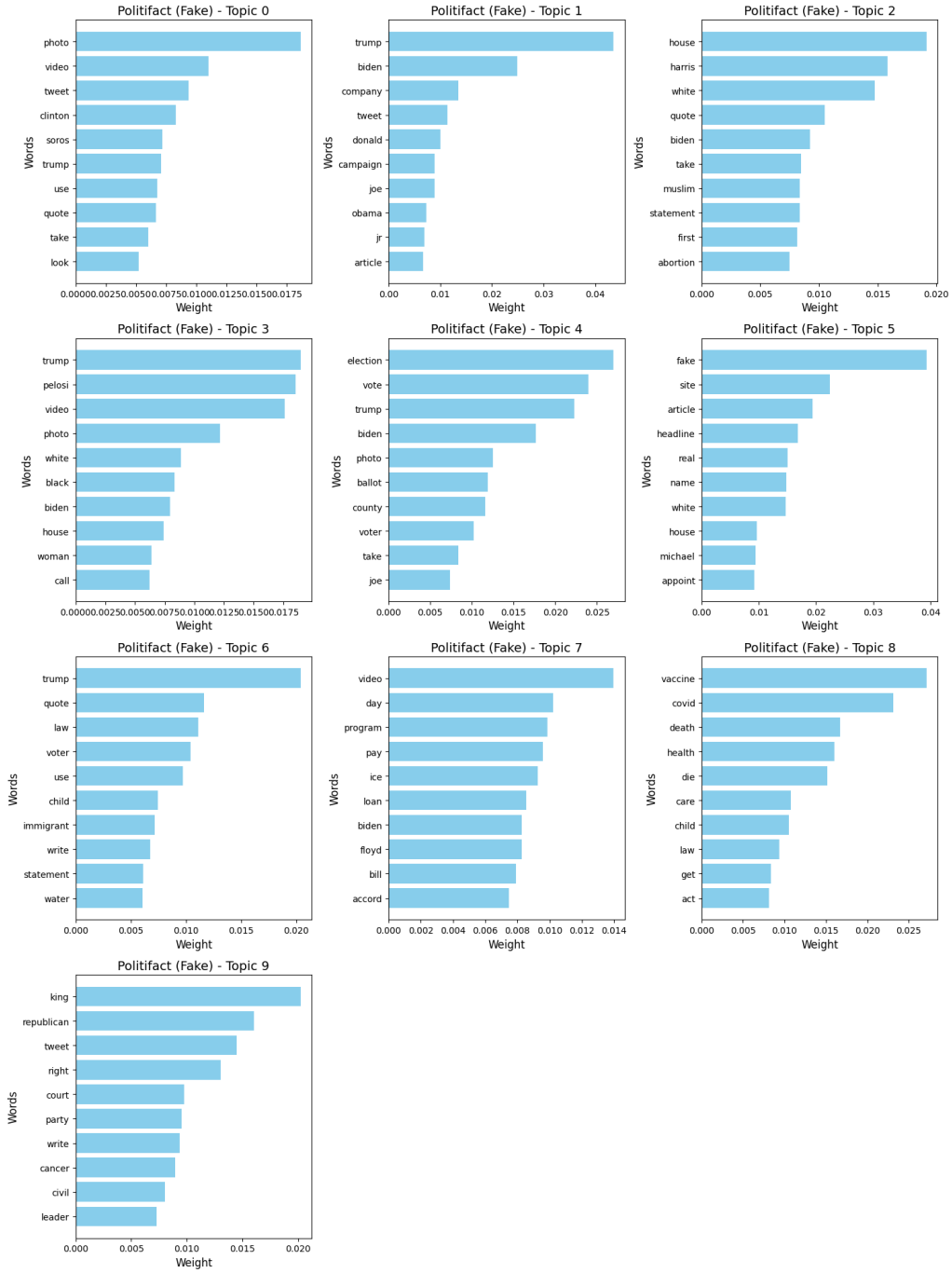



**Figure 13** - Most occurring words per LDA topic in Politifact, False claims

| Topic Number | Topic Name | Explanation | Document Count |
|---|---|---|---|
| Topic 0 | Misinformation Through Visual | This topic focuses on the spread of | 186 |

| | | | |
|---|---|---|---|
| | Media and Social Media Posts | misinformation via visual media (photos, videos) and social media posts, particularly involving political figures like Clinton, Soros, and Trump. | |
| Topic 1 | Political Figures and Campaign Narratives | This topic revolves around discussions related to political campaigns, companies, and tweets involving figures like Trump, Biden, and Obama. | 49 |
| Topic 2 | Identity Politics and Social Issues | Focuses on issues related to race, religion, gender, and policies such as abortion, often involving figures like Kamala Harris and Biden. | 35 |
| Topic 3 | Election Integrity and Political Manipulation | Discusses election-related misinformation, particularly about prominent politicians like Trump, Pelosi, and Biden, with emphasis on racial narratives. | 71 |
| Topic 4 | Voting and Ballot-Related Misinformation | Focuses on misinformation surrounding elections, voting processes, ballot fraud claims, and the involvement of candidates like Biden and Trump. | 49 |

| | | | |
|---|---|---|---|
| Topic 5 | Fake News and Media Manipulation | Covers misinformation spread by fake news websites, misleading headlines, and falsely attributed statements to political figures. | 64 |
| Topic 6 | Legal and Immigration Misinformation | Deals with legal policies, immigration narratives, and child-related policies, particularly in connection to Trump. | 51 |
| Topic 7 | Government Aid, Social Policies, and Scandals | Focuses on misinformation related to government aid programs, financial policies, and cases like George Floyd. | 30 |
| Topic 8 | Vaccine and COVID-19 Misinformation | Covers misinformation around COVID-19, vaccines, public health policies, and children's health concerns. | 33 |
| Topic 9 | Conservative and Right-Wing Narratives | Highlights narratives promoted within right-wing circles, including discussions on courts, civil rights, and political leadership. | 43 |

**Table 13** - Description of each LDA topic in Politifact, False claims